

# Troubleshooting Tools

**Task 1 :-** Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

**ANS :-**

```
IPv4 Address. . . . . : 10.86.106.47
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1c3a:5cff:fea7:37a8%2
                           10.86.106.3
```

**Task 2 :- Use the ping command to check the connectivity between your computer and [www.google.com](http://www.google.com), then note down the average response time shown.**

**ANS :-**

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asvin>ping www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 32 bytes of data:
Reply from 2001:4860:482c:7700::: time=55ms
Reply from 2001:4860:482c:7700::: time=38ms
Reply from 2001:4860:482c:7700::: time=52ms
Reply from 2001:4860:482c:7700::: time=43ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 55ms, Average = 47ms

C:\Users\asvin>
```

**Task 3 :- Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.**

**ANS :-**

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asvin>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2600:140f:7200:49::17d4:70]
over a maximum of 30 hops:

  1  34 ms    3 ms     2 ms  2409:40c1:4025:2267::b1
  2  *         *        *    Request timed out.
  3  41 ms    58 ms    24 ms  2405:200:5210:3:3925::1
  4  *         *        *    Request timed out.
  5  *         *        *    Request timed out.
  6  60 ms    23 ms    45 ms  2405:200:801:b00::39e
  7  *         *        *    Request timed out.
  8  234 ms   45 ms    43 ms  2405:203:400:100:172:31:3:23
  9  219 ms   42 ms    43 ms  2405:200:1602:600:49:44:187:bd
 10  60 ms    25 ms    36 ms  ae2.r02.stem01.bom01.fab.netarch.akamai.com [2600:140f:7600:303::1]
 11  70 ms    50 ms    28 ms  ae1.r03.spine02.bom01.fab.netarch.akamai.com [2600:140f:7600:e01::1]
 12  50 ms    73 ms    39 ms  ae3.r02.leaf01.bom01.fab.netarch.akamai.com [2600:140f:7600:e02::1]
 13  49 ms    32 ms    48 ms  ae2.r13.tor01.bom01.fab.netarch.akamai.com [2600:140f:7600:50d::1]
 14  48 ms    31 ms    56 ms  g2600-140f-7200-0049-0000-0000-17d4-0070.deploy.static.akamaitechnologies.com [2600:140f:7200:49::17d4:70]

Trace complete.

C:\Users\asvin>
```

**Task 4 :- Use the nslookup tool to find the IP address of [www.spotify.com](https://www.spotify.com) and explain in one line what this tool helps you verify.**

**ANS :-**

```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\asvin>nslookup www.spotify.com
Server: UnKnown
Address: 10.86.106.3

Non-authoritative answer:
Name: atc.spotify.map.fastly.net
Addresses: 2a04:4e42:31::810
           151.101.211.42
Aliases: www.spotify.com

C:\Users\asvin>
```

**What it verifies :**

nslookup verifies whether your DNS server can successfully convert a domain name (like [[www.spotify.com](https://www.spotify.com)](<https://www.spotify.com>)) into its corresponding IP address.

**Task 5 :- A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.**

**ANS :-**

**1. Ipconfig :-**

- Checks: If your laptop is properly connected to your home Wi-Fi and got a valid local IP address.

**2. nslookup zomato.com :-**

- Checks: If DNS is working—i.e., whether your internet can translate "zomato.com" into a numerical IP address.

**3. ping zomato.com :-**

- Checks: If your laptop can actually send and receive data packets to Zomato's server without dropping signal.